

Chapter 5

Constrained Optimization

Many (most) engineering optimisation problems are subject to *constraints*. For example, any problem involving voltages and currents will have maximum voltages that the circuit can tolerate, maximum currents due to wire material and thickness, etc. Constraints may be physical limitations, or even equations which guarantee that the design does what it is supposed. E.g. if we are optimizing an electric vehicle design, we will likely have constraints on features we want, e.g. the range of the car should be at least 200 km, or 300 km, or whatever we want.

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5.1 Introduction

This chapter focuses on *constrained* optimization problems which can be expressed as

$$\begin{aligned} \min f(\mathbf{x}) \\ \text{subject to } \underline{x}_i \leq x_i \leq \bar{x}_i, i = 1, 2, \dots, n \\ g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, n_g \\ h_l(\mathbf{x}) = 0, l = 1, 2, \dots, n_h. \end{aligned} \tag{5.1}$$

where n_g denotes the number of inequality constraints while $n_h \leq n$ represents the number of equality constraints.

The optimality conditions for constrained optimization problems are not as straightforward as those for unconstrained optimization. We begin with equality constraints because the mathematics and intuition are simpler, then add inequality constraints.

Both equality and inequality constraints state a condition that must be satisfied in order for the problem to be solved. If it is not possible to satisfy these constraints, the optimisation problem is infeasible. Equality constraints can be thought of as constraining the solution to lie on a line or plane in n -space (in a problem with 2 design variables and one equality constraint, it is exactly a line; whereas if there are 3 decision variables and one equality constraint, the solution must lie on a plane). Inequality constraints define a region in which the solution must lie. Therefore, the optimal value of the objective function can never improve (i.e. decrease, since we have formulated a minimization problem) via adding constraints.

5.2 Equality constraints

Let us consider the problem (5.1) with $n, n_g = 0$ i.e. only equality constraints. From the previous section, recall that $\nabla f(\mathbf{x})^T \mathbf{p} \geq 0$ was a necessary condition for optimality, i.e. in all directions the objective function does not improve. Of course, a local minimum $\mathbf{x}$ as opposed to the global minimum $\mathbf{x}^*$ would satisfy this condition as well, so this is not a sufficient condition for optimality.

What happens if we have equality constraints? We still need $\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0$ but now $\mathbf{p}$ is not every possible direction, but only *feasible* directions. Why? It does not matter if the objective function improves if we are barred from that region via an equality constraint.

How do we find feasible directions $\mathbf{p}$? We use the same Taylor-series approach:

$$h_j(\mathbf{x} + \mathbf{p}) \approx h_j(\mathbf{x}) + \nabla h_j(\mathbf{x})^T \mathbf{p}, l = 1, 2, \dots, n_h. \quad (5.2)$$

Assuming that $\mathbf{x}$ is a feasible point, then $h_j(\mathbf{x}) = 0$ for all constraints

$$h_j(\mathbf{x} + \mathbf{p}) \approx \nabla h_j(\mathbf{x})^T \mathbf{p}, l = 1, 2, \dots, n_h. \quad (5.3)$$

To remain feasible, $h_j(\mathbf{x} + \mathbf{p}) = 0$ for all j . We can write this in matrix form using the Jacobian matrix J_h as:

$$J_h(\mathbf{x})\mathbf{p} = 0. \quad (5.4)$$

This equation states that any feasible direction has to lie in the nullspace of the Jacobian of the constraints. What does this mean? Assuming that the constraints are linearly independent, J_h has full row rank and thus reduces the feasible space. In fact, for optimisation to be possible, $n > n_h$ (otherwise the feasible space reduces to a single point). See Fig. 5.8 in the textbook for illustration.

Having worked out our feasible directions, we now have that for $\mathbf{x}^*$ to be a constrained optimum

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0 \text{ for all } \mathbf{p} \text{ such that } J_h(\mathbf{x}^*)\mathbf{p} = 0. \quad (5.5)$$

Now, if $\mathbf{p}$ is a feasible direction, so is $-\mathbf{p}$ (think of a line - if we can go in one direction, we can also go in the other). Therefore, we require

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} = 0 \text{ for all } \mathbf{p} \text{ such that } J_h(\mathbf{x}^*)\mathbf{p} = 0. \quad (5.6)$$

Now, we have two equations, $\nabla f(\mathbf{x}^*)^T \mathbf{p} = 0$ and $J_h(\mathbf{x}^*)\mathbf{p} = 0$ which obviously bear resemblance to each other. This has a mathematical interpretation. Mathematically, (5.2) shows that p is in the nullspace of J_h . The row space of a matrix contains all the vectors that are orthogonal to its nullspace, and the rows of J_h are gradients of the constraints. Thus, the objective function gradient must be a linear combination of the gradients of the constraints.

(5.2) can thus be written as:

$$\nabla f(\mathbf{x}^*) = - \sum_{j=1}^{n_h} \lambda_j \nabla h_j(\mathbf{x}^*) \quad (5.7)$$

where λ_j are called the *Lagrange multipliers*. In matrix notation, we therefore have:

$$\begin{aligned} \nabla f(\mathbf{x}) &= -J_h(\mathbf{x})\lambda \\ h(\mathbf{x}) &= 0 \end{aligned} \quad (5.8)$$

In constrained optimization, it is sometimes convenient to use the Lagrangian function, which is a scalar function defined as

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + h(\mathbf{x})^T \lambda. \quad (5.9)$$

Taking the gradient of $\mathcal{L}$ with respect to both $\mathbf{x}$ and λ and setting them to zero yields (5.8). We have therefore transformed an n dimension constrained optimisation problem to an $n + n_h$ dimension unconstrained problem. However, this is not usually a good way to solve constrained optimization problems.

The optimality conditions just described are first-order conditions that are necessary but not sufficient. To make sure that a point is a constrained minimum, we also need to satisfy second-order conditions. For the unconstrained case, the Hessian of the objective function has to be positive definite. This is also sufficient for the constrained case, but we can get a less conservative condition since we only need to check the Hessian of the Lagrangian in the space of feasible directions. We therefore require the Hessian of the Lagrangian function to be positive for any feasible direction p .

Do [Example 5.2](#) and [Example 5.3](#) in the textbook.

5.3 Inequality constraints

We will re-use many of the same concepts, but must recall some terminology to begin.

- A constraint is *active* if $g_j(\mathbf{x}^*) = 0$.
- A constraint is *inactive* if $g_j(\mathbf{x}^*) < 0$.
- A constraint is *feasible* when $g_j(\mathbf{x}^*) \leq 0$.

As before, if x^* is an optimum, any small enough feasible step $\mathbf{p}$ from the optimum must result in a function increase. Based on Taylor series expansion, we have the condition:

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0 \quad (5.10)$$

which is the same as for the equality constrained case.

To consider inequality constraints, we use the same linearization as the equality constraints (5.2), but now we enforce an inequality to get:

$$g_j(\mathbf{x} + \mathbf{p}) \approx g_j(\mathbf{x}) + \nabla g_j(\mathbf{x})^T \mathbf{p} \leq 0, j = 1, 2, \dots, n_g. \quad (5.11)$$

For a given candidate point that satisfies all constraints, there are two possibilities to consider for each inequality constraint: whether the constraint is inactive ($g_j(\mathbf{x}^*) < 0$) or active ($g_j(\mathbf{x}^*) = 0$). If a given constraint is inactive, we do not need to add any condition for it because we can take a step $\mathbf{p}$ in any direction and remain feasible as long as the step is small enough. Thus, we only need to consider the active constraints for the optimality conditions.

For the equality constraint, we found that all first-order feasible directions are in the nullspace of the Jacobian matrix. Inequality constraints are not as restrictive. From (5.3), if constraint j is active ($g_j(\mathbf{x}) = 0$), then the nearby point $g_j(\mathbf{x} + \mathbf{p})$ is only feasible if $\nabla g_j(\mathbf{x})^T \mathbf{p} \leq 0$ for all constraints j that are active. In matrix form, we can write $J_g(\mathbf{x})\mathbf{p} \leq 0$, where the Jacobian matrix includes only the gradients of the active constraints.

The set of feasible directions that satisfies all active constraints is the intersection of all the closed half-spaces defined by the inequality constraints, that is, all $\mathbf{p}$ such that $J_g(\mathbf{x})\mathbf{p} \leq 0$.

This intersection of the feasible directions forms a polyhedral cone.

To find the cone of feasible directions, let us first consider the cone formed by the active inequality constraint gradients. This cone is defined by all vectors $\mathbf{d}$ such that:

$$\mathbf{d} = J_g^T \sigma = \sum_{j=1}^{n_g} \sigma_j \nabla g_j, \text{ where } \sigma_j \geq 0. \quad (5.12)$$

A direction $\mathbf{p}$ is feasible if $\mathbf{p}^T \mathbf{d} \leq 0$ for all $\mathbf{d}$ in the cone. The set of all feasible directions forms the polar cone of the cone defined by (5.3).

5.3.1 Farkas' Lemma

Now that we have established some intuition about the feasible directions, we need to establish under which conditions there is no feasible descent direction (i.e., we have reached an optimum). In other words, when is there no intersection between the cone of feasible directions and the open half-space of descent directions? To answer this question, we can use Farkas' lemma. This lemma states that given a rectangular matrix A with size $m \times n$ and a vector $\mathbf{b}$ with length m (i.e. the same length as the rows of the matrix), one (and only one) of two possibilities occurs:

1. There exists an $m \times 1$ vector $\mathbf{y}$ such that $A^T \mathbf{y} \geq 0$ and $\mathbf{b}^T \mathbf{y} < 0$.
2. There exists an $n \times 1$ vector $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq 0$.

In our case, replace A by the rectangular (i.e. $m = n$) matrix J_g^T , $\mathbf{b}$ by $-\nabla f$, $\mathbf{x}$ by σ , and $\mathbf{y}$ by $-\mathbf{p}$, and we have the following two possibilities:

1. There exists an $n \times 1$ vector $\mathbf{p}$ such that $J_g \mathbf{p} \leq 0$ and $\nabla f^T \mathbf{p} < 0$.
2. There exists an $n \times 1$ vector σ such that $J_g^T \sigma = \nabla f$ and $\sigma \geq 0$.

In the first case, this means there is a feasible direction $J_g(\mathbf{x}^*)\mathbf{p} \leq 0$ for which we can descend $\nabla f(\mathbf{x}^*)^T \mathbf{p} < 0$. In the second case, we can rearrange to state our optimality criterion:

$$\nabla f(\mathbf{x}^*) + J_g(\mathbf{x}^*)^T \sigma = 0, \text{ with } \sigma \geq 0. \quad (5.13)$$

This is similar to the conditions (5.8) for equality constraints. σ corresponds to the Lagrange multipliers for the inequality constraints and carries the additional restriction that $\sigma \geq 0$.

5.3.2 Slackness

Similar to the equality constrained case, we can construct a Lagrangian function whose stationary points are candidates for optimal points. We need to include all inequality constraints in the optimality conditions because we do not know in advance which constraints are active. To represent inequality constraints in the Lagrangian, we replace them with the equality constraints defined by:

$$g_j + s_j^2 = 0, j = 1, \dots, n_g. \quad (5.14)$$

where s_j is a new unknown associated with each inequality constraint called a *slack variable*. The slack variable is squared to ensure it is nonnegative, i.e. if (5.14) then $g_j \leq 0$ and the inequality constraint j is feasible.

The Lagrangian including both equality and inequality constraints is then

$$\mathcal{L}(\mathbf{x}, \lambda, \sigma, \mathbf{s}) = f(\mathbf{x}) + \lambda^T h(\mathbf{x}) + \sigma^T (g(\mathbf{x}) + \mathbf{s} \odot \mathbf{s}) \quad (5.15)$$

where σ represents the Lagrange multipliers associated with the inequality constraints. Here, we use $\odot$ to represent the element-wise multiplication of $\mathbf{s}$.

5.4 KKT conditions

Similar to the equality constrained case, we seek a stationary point for the Lagrangian, but now we have additional unknowns: the inequality Lagrange multipliers and the slack variables. Taking partial derivatives of the Lagrangian with respect to each set of unknowns and setting those derivatives to zero yields the first-order optimality conditions:

$$\nabla_x \mathcal{L} = 0 \implies \frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f}{\partial x_i} + \sum_{l=1}^{n_h} \lambda_l \frac{\partial h_l}{\partial x_i} + \sum_{j=1}^{n_g} \sigma_j \frac{\partial g_j}{\partial x_i} = 0, i = 1, \dots, n_x. \quad (5.16)$$

This criterion is the same as before but with additional Lagrange multipliers and constraints. Taking the derivatives with respect to the equality Lagrange multipliers, we have:

$$\nabla_\lambda \mathcal{L} = 0 \implies \frac{\partial \mathcal{L}}{\partial \lambda_l} = h_l = 0, l = 1, \dots, n_h. \quad (5.17)$$

which enforces the equality constraints as before. Taking derivatives with respect to the inequality Lagrange multipliers, we get:

$$\nabla_\sigma \mathcal{L} = 0 \implies \frac{\partial \mathcal{L}}{\partial \sigma_j} = g_j + s_j^2 = 0, j = 1, \dots, n_g. \quad (5.18)$$

which enforces the inequality constraints. Finally, differentiating the Lagrangian with respect to the slack variables, we obtain:

$$\nabla_s \mathcal{L} = 0 \implies \frac{\partial \mathcal{L}}{\partial s_j} = 2\sigma_j s_j = 0, j = 1, \dots, n_g. \quad (5.19)$$

which is called the complementary slackness condition. This condition helps us to distinguish the active constraints from the inactive ones. For each inequality constraint, either the Lagrange multiplier is zero (which means that the constraint is inactive), or the slack variable is zero (which means that the constraint is active).

Unfortunately, the complementary slackness condition introduces a combinatorial problem. The complexity of this problem grows exponentially with the number of inequality constraints because the number of possible combinations of active versus inactive constraints is 2^{n_g} .

In addition to the conditions for a stationary point of the Lagrangian (5.16)-(5.19), recall that we require the Lagrange multipliers for the active constraints to be nonnegative. Putting all these conditions together in matrix form, the first-order constrained

optimality conditions are as follows:

$$\nabla_f + J_h^T \lambda + J_g^T \sigma = 0 \quad (5.20a)$$

$$h = 0 \quad (5.20b)$$

$$g + s \odot s = 0 \quad (5.20c)$$

$$\sigma \odot s = 0 \quad (5.20d)$$

$$\sigma \geq 0 \quad (5.20e)$$

As in the equality constrained case, these first-order conditions are necessary but not sufficient. The second-order sufficient conditions require that the Hessian of the Lagrangian must be positive definite in all feasible directions, that is,

$$\mathbf{p}^T H_\lambda \mathbf{p} > 0 \quad (5.21a)$$

$$J_h \mathbf{p} = 0 \quad (5.21b)$$

$$J_g \mathbf{p} \leq 0 \quad (5.21c)$$

In other words, we only require positive definiteness in the intersection of the nullspace of the equality constraint Jacobian with the feasibility cone of the active inequality constraints.

Do [Example 5.4](#) and [Example 5.5](#) in the textbook.

Although these examples can be solved analytically, this is often difficult or impractical. The KKT conditions quickly become challenging to solve analytically, and as the number of constraints increases, trying all combinations of active and inactive constraints becomes intractable. Furthermore, engineering problems usually involve functions defined by models with implicit equations, which are impossible to solve analytically. The reason we include these analytic examples is to gain a better understanding of the KKT conditions. For the rest of the section, we focus on numerical methods, which are necessary for the vast majority of practical problems.

5.5 Lagrange multipliers again

The Lagrange multipliers quantify how much the corresponding constraints drive the design. More specifically, a Lagrange multiplier quantifies the sensitivity of the optimal objective function value $f(\mathbf{x}^*)$ to a variation in the value of the corresponding constraint. Here we explain why that is the case. We discuss only inequality constraints, but the same analysis applies to equality constraints.

When a constraint is inactive, the corresponding Lagrange multiplier is zero. This indicates that changing the value of an inactive constraint does not affect the optimum, as expected. This is only valid to the first order because the KKT conditions are based on the linearization of the objective and constraint functions. Because small changes are assumed in the linearization, we do not consider the case where an inactive constraint becomes active after perturbation.

It can be derived (see [Section 5.3.3](#) of the textbook) that:

$$\sigma_i = -\frac{df}{dg_i} \quad (5.22)$$

Thus, the Lagrange multipliers can predict how much improvement can be expected if a given constraint is relaxed. For inequality constraints, because the Lagrange multipliers

are positive at an optimum, this equation correctly predicts a decrease in the objective function value when the constraint value is increased.

The derivative defined in (5.22) has practical value because it tells us how much a given constraint drives the design. In this interpretation of the Lagrange multipliers, we need to consider the scaling of the problem and the units. Still, for similar quantities, they quantify the relative importance of the constraints.

5.6 Algorithms for solving constrained optimization problems

In general, if we can solve the KKT conditions, we have solved the optimization problem. However, as mentioned previous, there is a combinatorial issue with directly solving the KKT conditions for many problems, so we need an algorithm.

There are several methods used to solve constrained optimisation problems, however the most important solvers are the *interior point* and *SQP* methods. We do not have time to describe any method in detail, but will give a quick introduction here to interior point methods. We start by discussing the fundamental concept of penalty methods which underpin interior point solvers.

5.6.1 Penalty methods

The concept behind penalty methods is intuitive: to transform a constrained problem into an unconstrained one by adding a penalty to the objective function when constraints are violated or close to being violated. As mentioned in the introduction to this chapter, penalty methods are no longer used directly in gradient-based optimization algorithms because they have difficulty converging to the true solution. However, these methods are still valuable because, among other reasons, (1) they are simple and thus ease the transition into understanding constrained optimization; (2) penalty concepts are used in interior points methods. The penalized function can be written as:

$$\hat{f}(\mathbf{x}) = f(\mathbf{x}) + \mu\pi(\mathbf{x}) \quad (5.23)$$

where $\pi(\mathbf{x})$ is a penalty function, and the scalar μ is a penalty parameter. This is similar in form to the Lagrangian, but one difference is that μ is fixed instead of being a variable.

Various forms for $\pi(x)$ can be used, leading to different penalty methods. There are two main types of penalty functions: *exterior* penalties, which impose a penalty only when constraints are violated, and *interior* penalty functions, which impose a penalty that increases as a constraint is approached. The exterior penalty leads to slightly infeasible solutions, whereas an interior penalty leads to a feasible solution but underpredicts the objective.

We can use the unconstrained optimization techniques to minimize $\hat{f}(x)$. However, instead of just solving a single optimization problem, penalty methods usually solve a sequence of problems with different values of μ to get closer to the actual constrained minimum. The value of the penalty parameter μ must be chosen carefully. Mathematically, we recover the exact solution to the constrained problem only as μ tends to infinity. However, starting with a large value for μ is not practical. This is because the larger the value of μ the larger the Hessian condition number, which corresponds to the curvature

varying greatly with direction. This behavior makes the problem difficult to solve numerically. To solve the problem more effectively, we begin with a small value of μ and solve the unconstrained problem. We then increase μ and solve the new unconstrained problem, using the previous solution as the starting point. We repeat this process until the optimality conditions (or some other approximate convergence criteria) are satisfied.

By gradually increasing μ and reusing the solution from the previous problem, we avoid some of the ill-conditioning issues. Thus, the original constrained problem is transformed into a sequence of unconstrained optimization problems.

5.6.2 Interior point solvers

These methods form an objective similar to the interior penalty but with the key difference that instead of penalizing the constraints directly, they add slack variables to the set of optimization variables and penalize the slack variables. The resulting formulation is:

$$\begin{aligned} \min_{\mathbf{x}, s} \quad & f(\mathbf{x}) - \mu_b \sum_{j=1}^{n_g} \log s_j \\ \text{subject to} \quad & g(\mathbf{x}) + s = 0 \\ & h(\mathbf{x}) = 0 \end{aligned} \tag{5.24}$$

Instead of solving the KKT conditions of the original problem, we solve the KKT conditions of the interior-point formulation (5.24). This results in a linear problem that can be easily solved (see Equation 5.100 in the textbook).

These slack variables in (5.24) do not need to be squared, as was done in deriving the KKT conditions, because the logarithm is only defined for positive s values and acts as a barrier preventing negative values of s .

Similarly to penalty methods, we have to move slowly to prevent issues with conditioning due to the gradients. Determining the correct step-size is a crucial part of any interior-point algorithm, and it should be noted that this is used to enforce $s > 0$ implicitly. Hence, at each iteration, the parameters (also the barrier coefficient μ_b) are updated until we converge.

Time prevents a discussion of convergence, or indeed further discussion of interior-point methods, but the textbook has an excellent and detailed description for those wanting to know more. See especially Algorithm 5.7 in the textbook.

Interior point solvers are common in practice and are implemented in multiple software packages, e.g. *fmincon* in MATLAB or SciPy, IPOPT, Knitro, etc. Note: for those interested to make the comparison with SQP, SQP methods are also available in MATLAB (*sqp* option in *fmincon*), SNOPT, Knitro, and other packages as well. Reading <https://au.mathworks.com/help/optim/ug/choosing-the-algorithm.html> may be helpful.